Exterior Tensor Product

Assume that $V_1 = \cdots = V_m = V$

Alternating Map

Def.: An m-linear map $f: \underbrace{V \times \cdots \times V}_{m \text{ copies}} \to U$

is alternating

$\Leftrightarrow \forall \underline{w} \in V$, and $i, j \in \{1, \cdots, m\}$,

$$f(\underline{v}_1, \cdots, \underbrace{\underline{w}}_{i^{th}}, \cdots, \underbrace{\underline{w}}_{j^{th}}, \cdots, \underline{v}_m) = \underline{0}$$

E.g.:

(1) $\times : \mathbb{R}^3 \times \mathbb{R}^3 \to \mathbb{R}^3$

$\times(\underline{w}, \underline{w}) := \underline{w} \times \underline{w} = \underline{0}$

(3) $\det : \underbrace{\mathbb{F}^n \times \mathbb{F}^n}_{n \text{ copies}} \to \mathbb{F}$

(MAT2040) $\det(\cdots | \underline{w} | \cdots | \underline{w} | \cdots) = 0$

Exterior Tensor Product

Def.: the mth exterior tensor product $\Lambda^m V$ of V

is: $\Lambda^m V = \underbrace{(V \otimes \cdots \otimes V)}_{m \text{ copies}} / Z$,

where $Z = \text{Span}\{\cdots \otimes \underline{w} \otimes \cdots \otimes \underline{w} \otimes \cdots\}$,

(?) $\underline{w} \in V$

$\underline{v}_1 \wedge \cdots \wedge \underline{v}_m := (\underline{v}_1 \otimes \cdots \otimes \underline{v}_m) + Z$

$\in (V \otimes \cdots \otimes V)/Z = \Lambda^m V$

E.g.:

$(m = 4)$ In $\Lambda^4 V$:

(1) $\underline{w} \wedge \underline{v} \wedge \underline{u} \wedge \underline{w} = \underbrace{\underline{w} \otimes \underline{v} \otimes \underline{u} \otimes \underline{w}}_{\in Z} + Z$

(Rule 1) $= \underline{0} + Z = \underline{0}_{\Lambda^4 V}$

(2) $(\underline{w} + \underline{u}) \wedge \underline{v} \wedge (\underline{w} + \underline{u}) \wedge \underline{v} \overset{(1)}{=} \underline{0}_{\Lambda^4 V}$

Expand:

$$(\underline{w}+\underline{u})\otimes\underline{v}\otimes(\underline{w}+\underline{u})\otimes\underline{v}'+Z=\underline{0}$$

$$\Downarrow$$

$$\left(\cancel{\underline{w}\otimes\underline{v}\otimes\underline{w}\otimes\underline{v}'\in Z}+\underline{w}\otimes\underline{v}\otimes\underline{u}\otimes\underline{v}'+\underline{u}\otimes\underline{v}\otimes\underline{w}\otimes\underline{v}'+\cancel{\underline{u}\otimes\underline{v}\otimes\underline{u}\otimes\underline{v}'}\right)+Z=\underline{0}$$

$$\Downarrow \qquad \in Z$$

$$\underline{w}\wedge\underline{v}\wedge\underline{u}\wedge\underline{v}'+\underline{u}\wedge\underline{v}\wedge\underline{w}\wedge\underline{v}'=\underline{0}$$

$$\Downarrow$$

$$\underline{w}\wedge\underline{v}\wedge\underline{u}\wedge\underline{v}'=-(\underline{u}\wedge\underline{v}\wedge\underline{w}\wedge\underline{v}')$$

(Rule 2) ←

(3) $V=\mathbb{R}^3=\mathrm{Span}\{\underline{e}_1,\underline{e}_2,\underline{e}_3\}$. In $\Lambda^2\mathbb{R}^3$

$$\begin{pmatrix}1\\2\\3\end{pmatrix}\wedge\begin{pmatrix}4\\5\\6\end{pmatrix}=(\underline{e}_1+2\underline{e}_2+3\underline{e}_3)\wedge(4\underline{e}_1+5\underline{e}_2+6\underline{e}_3)$$

$$=4\,\underline{e}_1\wedge\underline{e}_1+8\,\underline{e}_2\wedge\underline{e}_1+12\underline{e}_3\wedge\underline{e}_1+5\,\underline{e}_1\wedge\underline{e}_2+10\,\underline{e}_2\wedge\underline{e}_2+15\underline{e}_3\wedge\underline{e}_2+6\,\underline{e}_1\wedge\underline{e}_3+12\,\underline{e}_2\wedge\underline{e}_3+18\underline{e}_3\wedge\underline{e}_3$$

$$=(5-8)\,\underline{e}_1\wedge\underline{e}_2+(6-12)\underline{e}_1\wedge\underline{e}_3+(12-15)\underline{e}_2\wedge\underline{e}_3$$

Prop.: Let $\dim(V) = n$

$\{\underline{v}_1, \cdots, \underline{v}_n\}$ be a basis of V

Then

$\forall m \leq n$,

a basis of $\Lambda^m V$

is $\{\underline{v}_{i_1} \wedge \cdots \wedge \underline{v}_{i_m} \mid 1 \leq i_1 \leq \cdots \leq i_m \leq n\}$,

$$\dim(\Lambda^m V) = \binom{n}{m} = \frac{n!}{m!(n-m)!}$$

E.g. 1: $\dim(\Lambda^2 \mathbb{R}^3) = \binom{3}{2} = 3$

a basis of $\Lambda^2 \mathbb{R}^3$: $\{\underline{e}_1 \wedge \underline{e}_2, \underline{e}_1 \wedge \underline{e}_3, \underline{e}_2 \wedge \underline{e}_3\}$

E.g. 2:

Example Basis for

$\Lambda^1 \mathbb{R}^4$: $\{\underline{e}_1, \underline{e}_2, \underline{e}_3, \underline{e}_4\}$

$\Lambda^2 \mathbb{R}^4$: $\{\underline{e}_1 \wedge \underline{e}_2, \underline{e}_1 \wedge \underline{e}_3, \underline{e}_1 \wedge \underline{e}_4, \underline{e}_2 \wedge \underline{e}_3, \underline{e}_2 \wedge \underline{e}_4, \underline{e}_3 \wedge \underline{e}_4\}$

$\Lambda^3 \mathbb{R}^4$: $\{\underline{e}_1 \underline{e}_2 \underline{e}_3, \underline{e}_1 \underline{e}_3 \underline{e}_4, \underline{e}_1 \underline{e}_3 \underline{e}_4, \underline{e}_2 \underline{e}_3 \underline{e}_4\}$

$\Lambda^4 \mathbb{R}^4$: $\{\underline{e}_1 \wedge \underline{e}_2 \wedge \underline{e}_3 \wedge \underline{e}_4\}$ ← 1-dim'l

(總結:)

	$V_1 \otimes \cdots \otimes V_m$		$\Lambda^m V$	
General element	$\underline{v}_1^{(1)} \otimes \cdots \otimes \underline{v}_m^{(1)} + \cdots + \underline{v}_1^{(k)} \otimes \cdots \otimes \underline{v}_m^{(k)}$		$(\underline{v}_1^{(1)} \wedge \cdots \wedge \underline{v}_m^{(1)}) + \cdots + (\underline{v}_1^{(k)} \wedge \cdots \wedge \underline{v}_m^{(k)})$	
Dimension	$\dim(V_1) \cdots \dim(V_m)$		$\binom{n}{m} = \frac{n!}{m!(n-m)!}$	
Universal Property	$f: V_1 \times \cdots \times V_m \to U$ m-linear	$\rightsquigarrow$	$\Phi: V_1 \otimes \cdots \otimes V_m \to U$ linear trans	
	$f: V \times \cdots \times V \to U$ alternating	$\rightsquigarrow$	$\Phi: \Lambda^m V \to U$ linear trans	
Linear trans	$T_1: V_1 \to V_1'$ $\vdots$ $T_m: V_m \to V_m'$	$\rightsquigarrow$	$T_1 \otimes \cdots \otimes T_m$: $V_1 \otimes \cdots \otimes V_m \to V_1' \otimes \cdots \otimes V_m'$	
	$T_1: V \to V'$ $\vdots$ $T_m: V \to V'$	$\rightsquigarrow$	$T_1 \wedge \cdots \wedge T_m$: $\Lambda^m V \to \Lambda^m V$	

Note: the universal property and construction of linear trans. holds for $\wedge^m V$,

i.e. $T_i : V \to V'$, $i \in \{1, \cdots, m\}$

$\Downarrow$

$T_1 \wedge \cdots \wedge T_m : \wedge^m V \to \wedge^m V'$,

s.t. $(T_1 \wedge \cdots \wedge T_m)(\underline{v}_1 \wedge \cdots \wedge \underline{v}_m) = T(\underline{v}_1) \wedge \cdots \wedge T(\underline{v}_m)$

~~不自~~ ⓒ

The determinant

Def.: let $\dim(V) = n$,

$T: V \to V$ be a linear operator

Consider $\underbrace{T \wedge \cdots \wedge T}_{n \text{ copies}} : \underbrace{\wedge^n V}_{\dim(\wedge^n V) = \binom{n}{n} = 1} \to \wedge^n V$ $= \underline{\text{Span}(\underline{\alpha})}$?

given by

$T \wedge \cdots \wedge T(\alpha) = \blacktriangle \underline{\alpha}$ by ⓒ some ? scalar $\blacktriangle \in \mathbb{F}$

Define $\det(T) := \blacktriangle$

E.g.?

Example $V=\mathbb{R}^2$ $T(\underline{x})=\begin{pmatrix} a & b \\ c & d \end{pmatrix}\underline{x}$

What is $\Lambda^2\mathbb{R}^2$? This is $\Lambda^2\mathbb{R}^2 = \mathrm{Span}\{\underline{e}_1\wedge\underline{e}_2\}$ ("$\mathbb{R}$")

What is $T\wedge T =: T^{\wedge 2}: \Lambda^2\mathbb{R}^2 \to \Lambda^2\mathbb{R}^2$?

$T^{\wedge 2}(\underline{\alpha}) = (T\wedge T)(\underline{e}_1\wedge\underline{e}_2)$

$= T(\underline{e}_1)\wedge T(\underline{e}_2) = \begin{pmatrix} a \\ c \end{pmatrix}\wedge\begin{pmatrix} b \\ d \end{pmatrix}$ $=$ $(a\underline{e}_1+c\underline{e}_2)\wedge(b\underline{e}_1+d\underline{e}_2)$

$=$ $ad\,\underline{e}_1\wedge\underline{e}_2 + bc\,\underline{e}_2\wedge\underline{e}_1$

$=(ad-bc)(\underline{e}_1\wedge\underline{e}_2)$

$\det(T)$ $=$ $ad-bc$

Properties (of $\det_n$):

Let $V=\mathbb{F}^n$, $\Lambda^n\mathbb{F}^n := \mathrm{Span}\{\underline{e}_1\wedge\cdots\wedge\underline{e}_n\}$

$\cong \mathbb{F}$ via $\underline{e}_1\wedge\cdots\wedge\underline{e}_n =: \alpha \mapsto 0$

Suppose $T:\mathbb{F}^n\to\mathbb{F}^n$,

(1) $T(\underline{x}) = \left(\cdots\middle|\underset{i\text{th}}{\underline{w}}\middle|\cdots\middle|\underset{j\text{th}}{\underline{w}}\middle|\cdots\right)\underline{x}$. Then $\det(T)=0$

$T^{\wedge n}(\underline{\alpha}) = T^{\wedge n}(\underline{e}_1\wedge\cdots\wedge\underline{e}_n)$

$= T(\underline{e}_1)\wedge\cdots\wedge T(\underline{e}_i)\wedge\cdots\wedge T(\underline{e}_j)\wedge\cdots\wedge T(\underline{e}_n)$

$= T(\underline{e}_1)\wedge\cdots\wedge\underline{w}\wedge\cdots\wedge\underline{w}\wedge\cdots\wedge T(\underline{e}_n) = \underline{0}\cdot\underline{\alpha}$

(2) $T(\underline{x}) = \left(\underline{v}_1\middle|\cdots\middle|\underline{v}_n\right)\underline{x}$

Then

$T^{\wedge n}(\underline{\alpha}) = T(\underline{e}_1)\wedge\cdots\wedge T(\underline{e}_i)\wedge\cdots\wedge T(\underline{e}_n)$

$= \underline{v}_1\wedge\cdots\wedge\underline{v}_i\wedge\cdots\wedge\underline{v}_n$

$= (\underline{v}_1)\wedge\cdots\wedge\left(k_1'\underline{v}_1 + \cdots + k_{i-1}'\underline{v}_{i-1} + k_{i+1}'\underline{v}_{i+1} + \cdots + k_n'\underline{v}_n\right)\wedge\cdots\wedge\underline{v}_n$

$\underline{v}_1,\ldots,\underline{v}_n$ are linearly dependent

e.g. $k_1\underline{v}_1+\cdots+k_i\underline{v}_i+\cdots+k_n\underline{v}_n=\underline{0}$ for some $k_i\neq 0$.

$\Rightarrow \underline{v}_i = \frac{-k_1}{k_i}\underline{v}_1+\cdots+\widehat{\underline{v}_i}+\cdots+\frac{-k_n}{k_i}\underline{v}_n$ (no $\underline{v}_i$ term)

$= \underline{0}+\underline{0}+\cdots+\underline{0} = 0\cdot\underline{\alpha} \Rightarrow \det(T)=0$

(3) $T(\underline{x}) = \left(\underline{v}_1 \,|\, \cdots \,|\, \underline{v}_i + \underline{v}_i' \,|\, \cdots \,|\, \underline{v}_n\right)\underline{x}$

$S(\underline{x}) = (\underline{v}_1 \,|\, \cdots \,|\, \underline{v}_i \,|\, \cdots \,|\, \underline{v}_n)\underline{x}$

$S'(\underline{x}) = (\underline{v}_1 \,|\, \cdots \,|\, \underline{v}_i' \,|\, \cdots \,|\, \underline{v}_n)\underline{x}$

Claim

$\det(T) = \det(S) + \det(S')$

$T^{\wedge n}(\underline{\alpha}) = \underline{v}_1 \wedge \cdots \wedge (\underline{v}_i + \underline{v}_i') \wedge \cdots \wedge \underline{v}_n$

$\det(T)\,\underline{\alpha} = (\underline{v}_1 \wedge \cdots \wedge \underline{v}_i \wedge \cdots \wedge \underline{v}_n) + (\underline{v}_1 \wedge \cdots \wedge \underline{v}_i' \wedge \cdots \wedge \underline{v}_n)$

$= \det(S)\,\underline{\alpha} + \det(S')\,\underline{\alpha}$

$= (\det(S) + \det(S'))\,\underline{\alpha}$

(4) $T(\underline{x}) = A\underline{x}$, $S(\underline{x}) = B\underline{x}$

$(T \circ S)(\underline{x}) = ABx$

Claim:

$\det(T \circ S)$ ✓

$= \det(T)\det(S)$

$\det(S) + \det(S')$

$\underline{\det(T \circ S)\,\underline{\alpha}} = (T \circ S)^{\wedge n}(\underline{\alpha}) = (T \circ S)^{\wedge n}(\underline{e}_1 \wedge \cdots \wedge \underline{e}_n)$

$= (T \circ S)(\underline{e}_1) \wedge \cdots \wedge (T \circ S)(\underline{e}_n)$

$= T^{\wedge n}\left(S(\underline{e}_1) \wedge \cdots \wedge S(\underline{e}_n)\right)$

$= T^{\wedge n} \circ \left(S^{\wedge n}(\underline{e}_1 \wedge \cdots \wedge \underline{e}_n)\right)$

$= T^{\wedge n} \circ S^{\wedge n}(\underline{\alpha})$

$= T^{\wedge n}(\det(S)\,\underline{\alpha})$

$= \det(S)\left(T^{\wedge n}(\underline{\alpha})\right)$

$= \det(S)\det(T)\,\underline{\alpha}$

(3) $T(\underline{x}) = (\underline{v}_1 | \cdots | \underline{v}_i + \underline{v}_i' | \cdots | \underline{v}_n)\underline{x}$

$S(\underline{x}) = (\underline{v}_1 | \cdots | \underline{v}_i | \cdots | \underline{v}_n)\underline{x}$

$S'(\underline{x}) = (\frown | \underline{v}_i' | \frown)\underline{x}$

(In MAT2040 : $\det(T) = \det(S) + \det(S')$)

$T^{\underline{1^n}}_{\underline{0}}(\underline{\alpha}) = \underline{v}_1 \wedge \cdots \wedge (\underline{v}_i$

(4) $T(\underline{x}) = A\underline{x}$,

$S(\underline{x}) = B\underline{x}$

$(T \circ S)(\underline{x}) = AB\underline{x}$

(~ : $\det(T \circ S) = \det(T) \cdot \det$